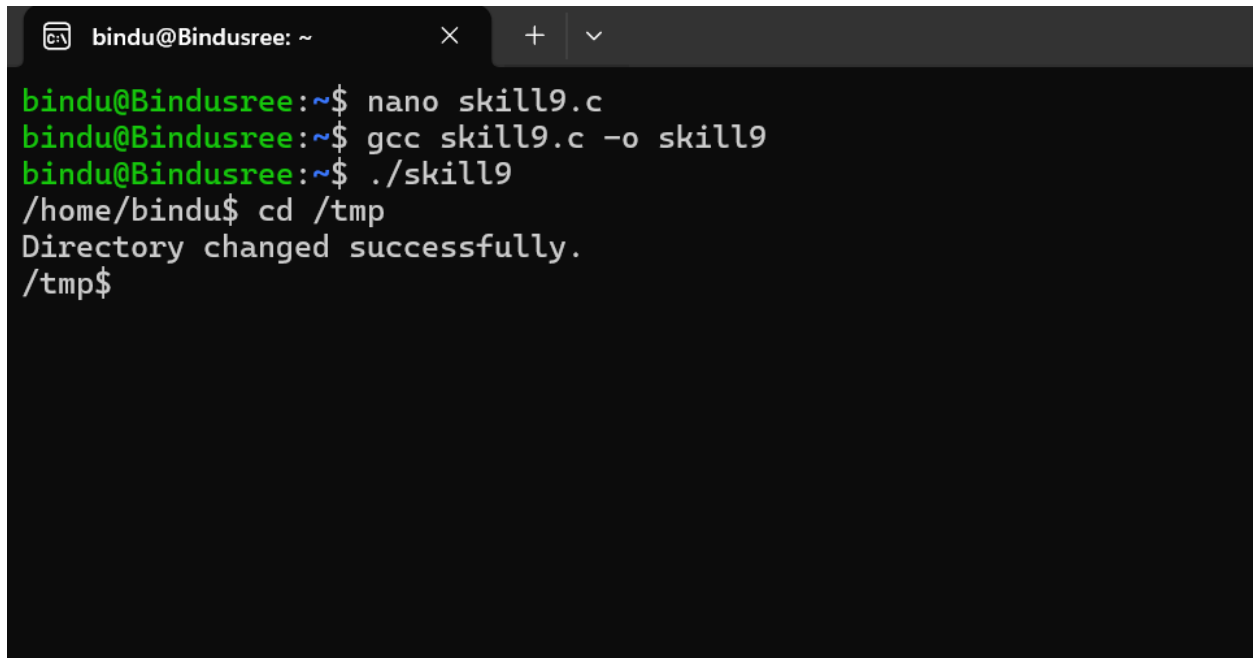


9.To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

A terminal window with a dark background and light green text. The window title bar shows 'bindu@Bindusree: ~' and standard window controls. The terminal displays a sequence of commands and their outputs: 'nano skill9.c' is entered; 'gcc skill9.c -o skill9' is entered; './skill9' is entered, followed by a new prompt '/home/bindu\$'; 'cd /tmp' is entered, followed by the output 'Directory changed successfully.' and a new prompt '/tmp\$'.

```
bindu@Bindusree:~$ nano skill9.c
bindu@Bindusree:~$ gcc skill9.c -o skill9
bindu@Bindusree:~$ ./skill9
/home/bindu$ cd /tmp
Directory changed successfully.
/tmp$
```

```

#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>
#include <limits.h>

int main() {
    char input[200];
    char previous[PATH_MAX] = "";

    while (1) {
        char current[PATH_MAX];

        getcwd(current, sizeof(current));
        printf("%s$ ", current);

        fgets(input, sizeof(input), stdin);
        input[strcspn(input, "\n")] = '\0';

        // Exit
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // cd command
        if (strncmp(input, "cd ", 3) == 0) {
            char *path = input + 3;

            // Save current directory
            strcpy(previous, current);

            if (chdir(path) == -1) {
                perror("cd failed");
            } else {
                printf("Directory changed successfully.\n");
            }
        }
    }
}

```

```

// cd -
else if (strcmp(input, "cd -") == 0) {
    if (strlen(previous) == 0) {
        printf("Previous directory not available.\n");
    } else {
        char temp[PATH_MAX];

        strcpy(temp, current);

        if (chdir(previous) == -1) {
            perror("cd failed");
        } else {
            strcpy(previous, temp);
            printf("Changed to previous directory.\n");
        }
    }
}

else {
    printf("Unknown command: %s\n", input);
}

return 0;

```